THE PICARD GROUP OF PROJECTIVE SPACE OVER A UNIQUE FACTORISATION DOMAIN IS THE INTEGERS, GENERATED BY THE TWISTING SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-Pic-projective-space-UFD is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 7254-7264 and proof span 7266-7301 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof works with the author's notation for the standard charts, the units, and the exponents of the coordinate ratios; the displays are preserved unchanged.

Editorial note. 1 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The Picard group of an open subscheme of a unique factorisation domain's affine space is trivial, cited from the same chapter. Cited at proof lines 7266-7268. Shows that the standard affine charts of projective space have trivial Picard group, so invertible modules are given by cocycles for that cover.
- The description of the units of a localization as the product of the units of the base with the powers of the localizing element, and the cocycle description of invertible modules with respect to a cover. Used at proof lines 7272-7288. Turns the cocycle into a pair of data, a unit-valued part and an integral exponent part, for each overlapping pair of charts.
- The trivializing section of a twist of the structure sheaf on a standard chart and the corresponding cocycle. Used at proof lines 7295-7301. Identifies the twisting sheaf with the class of a cocycle consisting of powers of the ratios of coordinates, matching the classification obtained.

Source reference. Original lemma 0BDA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14308–14308. Same-chapter reference used in the retained author text at source line(s) 7267. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's

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own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let R be a UFD. The Picard group of $\mathbf{P}^n_R$ is $\mathbf{Z}$. More precisely, there is an isomorphism*

$$\mathbf{Z} \longrightarrow \mathrm{Pic}(\mathbf{P}^n_R), \quad m \longmapsto \mathcal{O}_{\mathbf{P}^n_R}(m)$$

In particular, the Picard group of $\mathbf{P}^n_k$ of projective space over a field k is $\mathbf{Z}$.

Proof. By Lemma 0BDA the Picard groups of the opens $D_+(T_i) \cong \mathbf{A}^n_R$ are trivial. Thus if $\mathcal{L}$ is an invertible module on $\mathbf{P}^n_R$, then it is given by a 1-cocycle with values in the sheaf of invertible functors $\mathcal{O}^*$ for the open covering

$$\mathbf{P}^n_R = \bigcup\nolimits_{i=0,\ldots,n} D_+(T_i)$$

Observe that for $i \neq j$ we have

$$\mathcal{O}^*(D_+(T_i) \cap D_+(T_j)) = \mathcal{O}^*(D_+(T_iT_j)) = \left(R[T_0, \ldots, T_n]_{(T_iT_j)}\right)^* = R^* \times (T_i/T_j)^{\mathbf{Z}}$$

Thus such a cocycle (g_{ij}) is given by units

$$g_{ij} = u_{ij}(T_i/T_j)^{e_{ij}}$$

with $u_{ij} \in R^*$ and $e_{ij} \in \mathbf{Z}$ satisfying the cocycle condition. The cocycle condition over $D_+(T_iT_jT_k)$ for $\#\{i, j, k\} = 3$ tell us that

$$u_{ik} = u_{ij}u_{jk} \quad \text{and} \quad e_{ik} = e_{ij} = e_{jk}$$

Whence $u_{ij} = u_{i1}u_{j1}^{-1}$ is a boundary. Thus all isomorphism classes of invertible modules are given by taking the cocycle with

$$g_{ij} = (T_i/T_j)^e$$

for some $e \in \mathbf{Z}$. Since $\mathcal{O}(n)$ has trivializing section T_i^n over $D_+(T_i)$ we see that the corresponding cocycle of $\mathcal{O}(n)$ is $(T_i/T_j)^n$ and the proof is complete. □